Pico-IIC读取数据

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- 1.连接设备
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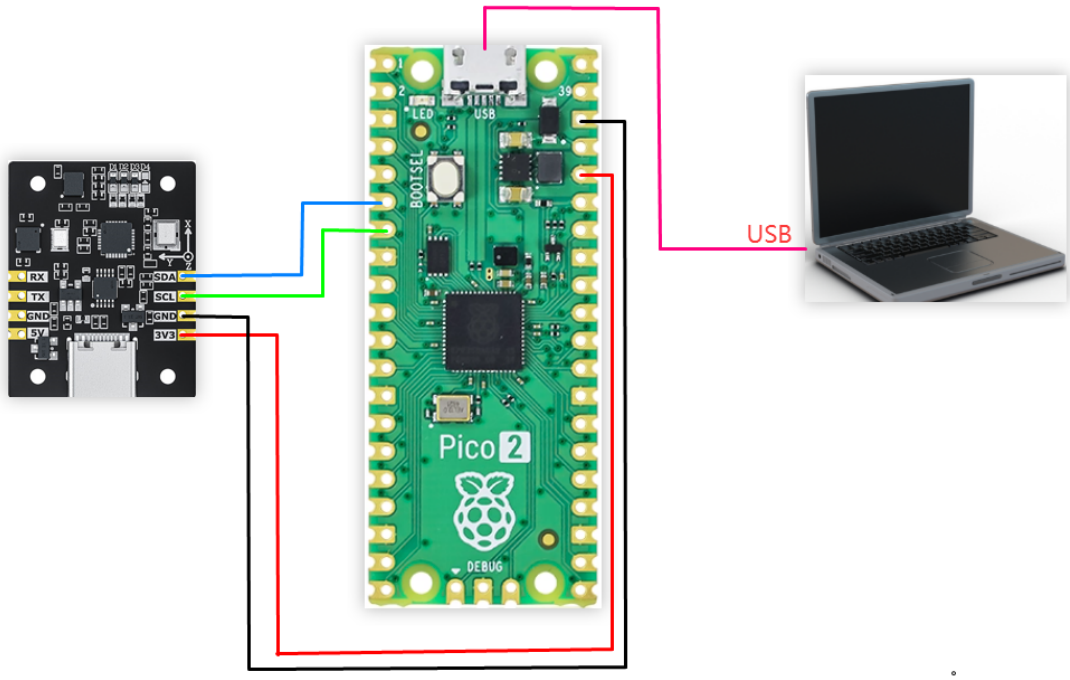
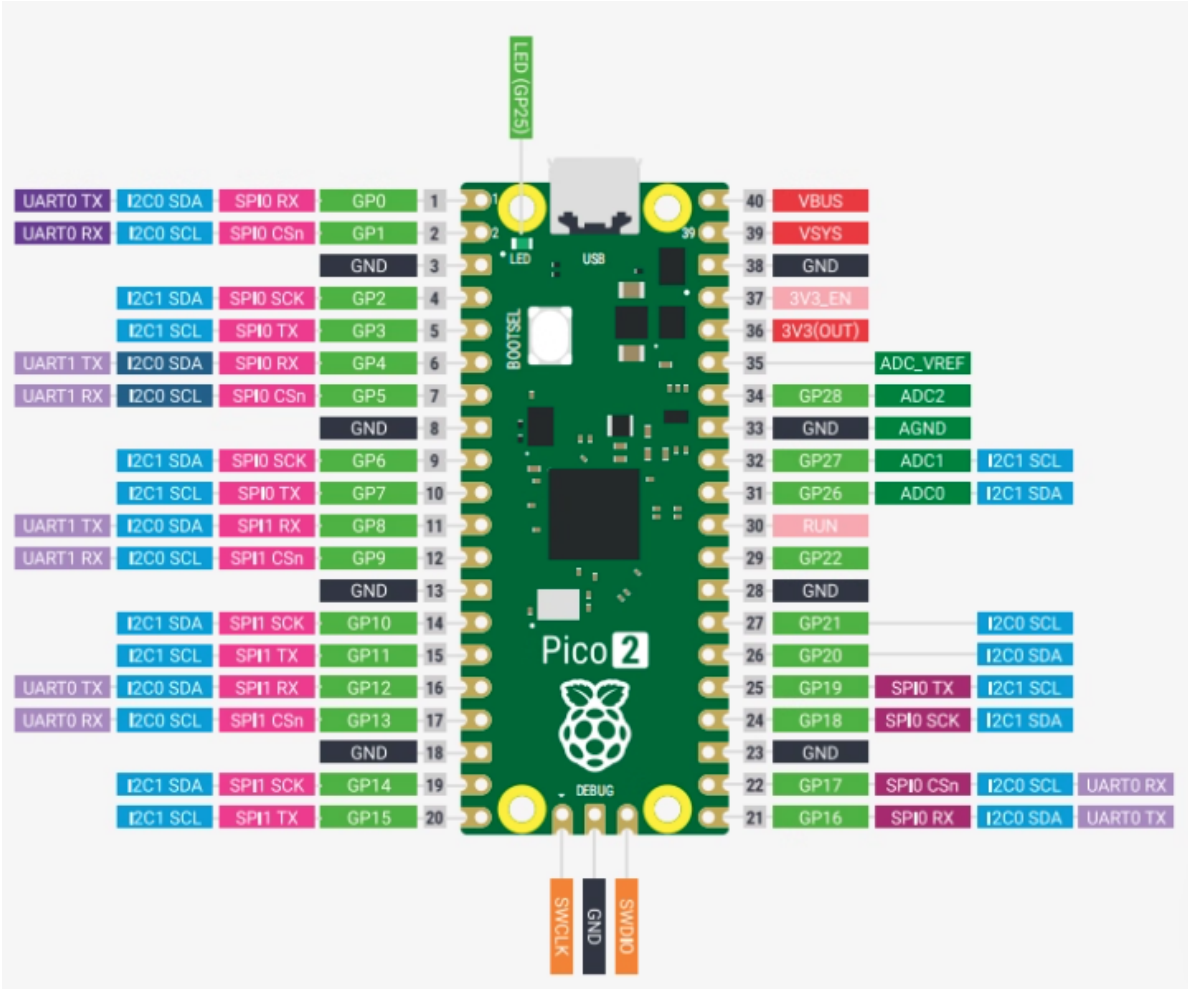
- 3.读取imu数据

本次例程使用的是Pico2开发板，一台windows电脑、杜邦线若干、IMU姿态传感器。

1.连接设备



IMU姿态传感器	Pico2
SDA	GP4
SCL	GP5
GND	GND
5V	3V3(OUT)



2.关键代码解析

具体代码请看资料中的源码。

```
# 获取加速度计三轴数据，返回accel=[a_x, a_y, a_z]
    # Get accelerometer triaxial data, return accel=[a_x, a_y, a_z]
    def get_accelerometer_data(self):
        values = self._read_data(self.FUNC_RAW_ACCEL, 6)
        # 转化单位为g
        accel_ratio = 16 / 32767.0
        a_x = struct.unpack('h', bytearray(values[0:2]))[0]*accel_ratio
        a_y = struct.unpack('h', bytearray(values[2:4]))[0]*accel_ratio
        a_z = struct.unpack('h', bytearray(values[4:6]))[0]*accel_ratio
        accel = [a_x, a_y, a_z]
        if self._delay_time > 0:
            time.sleep(self._delay_time)
        return accel

    # 获取陀螺仪三轴数据，返回gyro=[g_x, g_y, g_z]
    # Get the gyro triaxial data, return gyro=[g_x, g_y, g_z]
    def get_gyroscope_data(self):
        values = self._read_data(self.FUNC_RAW_GYRO, 6)
        # 转化单位为rad/s
        AtoR = math.pi / 180.0
        gyro_ratio = (2000 / 32767.0) * AtoR
        g_x = struct.unpack('h', bytearray(values[0:2]))[0]*gyro_ratio
        g_y = struct.unpack('h', bytearray(values[2:4]))[0]*gyro_ratio
        g_z = struct.unpack('h', bytearray(values[4:6]))[0]*gyro_ratio
        gyro = [g_x, g_y, g_z]
```

```
        if self._delay_time > 0:
            time.sleep(self._delay_time)
        return gyro


    # 获取IMU的四元数，返回quat=[w, x, y, z]
    def get_imu_quaternion_data(self):
        values = self._read_data(self.FUNC_QUAT, 16)
        q0 = struct.unpack('f', bytearray(values[0:4]))[0]
        q1 = struct.unpack('f', bytearray(values[4:8]))[0]
        q2 = struct.unpack('f', bytearray(values[8:12]))[0]
        q3 = struct.unpack('f', bytearray(values[12:16]))[0]
        quat = [q0, q1, q2, q3]
        if self._delay_time > 0:
            time.sleep(self._delay_time)
        return quat

    # 获取板子姿态角，返回euler=[roll, pitch, yaw]
    # ToAngle=True返回角度，ToAngle=False返回弧度。
    def get_imu_attitude_data(self, ToAngle=True):
        values = self._read_data(self.FUNC_EULER, 12)
        roll = struct.unpack('f', bytearray(values[0:4]))[0]
        pitch = struct.unpack('f', bytearray(values[4:8]))[0]
        yaw = struct.unpack('f', bytearray(values[8:12]))[0]
        if ToAngle:
            RtoA = 180.0 / math.pi
            roll = roll * RtoA
            pitch = pitch * RtoA
            yaw = yaw * RtoA
        euler = [roll, pitch, yaw]
        if self._delay_time > 0:
            time.sleep(self._delay_time)
        return euler
```

get_accelerometer_data():获取加速度计三轴数据，返回accel=[a_x, a_y, a_z]

get_gyroscope_data():获取陀螺仪三轴数据，返回gyro=[g_x, g_y, g_z]

get_imu_quaternion_data():获取IMU的四元数，返回quat=[w, x, y, z]

get_imu_attitude_data():获取板子姿态角，返回euler=[roll, pitch, yaw]

3.读取imu数据

程序运行后，可以在下面的终端显示imu的数据。

IIC.py

```python
from machine import I2C, Pin
import struct
import time
import math


READ_INTERVAL = 0.1  # 传感器读取间隔（秒） Sensor read interval in seconds


class YbImuI2c(object):

    def __init__(self, debug=False):

        # 创建I2C通信对象   Create I2C communication object
        self.dev = I2C(0, scl=Pin(5), sda=Pin(4), freq=400000)  # 使用I2C0，SCL=GP5，SDA=GP4
                                                                # Use I2C0, SCL=GP5, SDA=GP4

        self._delay_time = 0.001
        self._debug = debug
        self._addr = 0x23

        self.FUNC_VERSION = 0x01
```

Shell

```
------ Sensor Data ------
Acceleration [g]:       x=-0.914, y=-1.100, z= 0.721
Gyroscope [rad/s]:      x=-0.837, y= 1.670, z=-0.289
Magnetometer [uT]:      x=-43.751, y= 5.591, z=-30.348
Quaternion:             w=-0.07147, x= 0.36896, y= 0.35920, z=-0.85227
Euler Angle [deg]:      roll=-54.77, pitch= 35.28, yaw= 151.44
Barometer:              height= 0.25 m, temperature= 41.95 °C
                        pressure= 101479.05469 Pa, pressure_diff= 101482.06250 Pa
-------------------------
```

MicroPython (Raspberry Pi Pico)

注意：以上为10轴IMU的数据读取，6轴无磁力计（Magnetometer）与气压计（Barometer）数据，9轴无气压计（Barometer）数据。